

ES02 Aleph Carrier Board Data Sheet

April 2025

- Carrier board revision: **2.0.0**

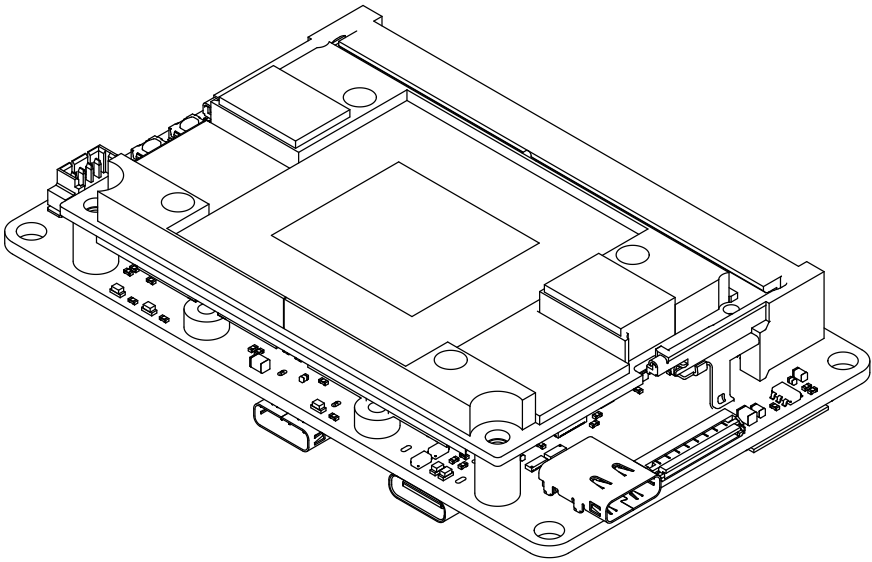


Figure 1: Aleph Carrier Board

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Overview

The Aleph Carrier Board is a space-grade flight computer designed to house and interface the NVIDIA Jetson Orin NX SOM series for high-performance, on-orbit compute. Aleph enables AI-based autonomy, real-time image processing, and advanced decision-making directly on-board, reducing ground station dependency and latency.

Key Features

- Peripheral support: 2× I2C, 1× UART, CAN, SPI, USB FS, USB SS+ (10 Gbps), Serial Debug, Gigabit Ethernet
- High-speed interfaces: PCIe x4, x2, x1; M.2 E-Key and M-Key
- Operational temperature range: -40°C to +85°C
- Compact mechanical design: 89 × 53.5 × 19.3 mm
- Modular and open software stack: Runs Elodin’s open-source flight software modules or fully user-defined mission software.
- All components and materials are selected for low outgassing.

Specifications

Environmental

Minimum operating temperature (°C)	-45
Maximum operating temperature (°C)	85
ROHS compliant	Yes
Vibration qualification	Please contact us for further information

Mechanical

Dimensions (mm)	Refer to Figure 2, 3, 4
Base Material	Isola 370HR
Surface finish	ENIG

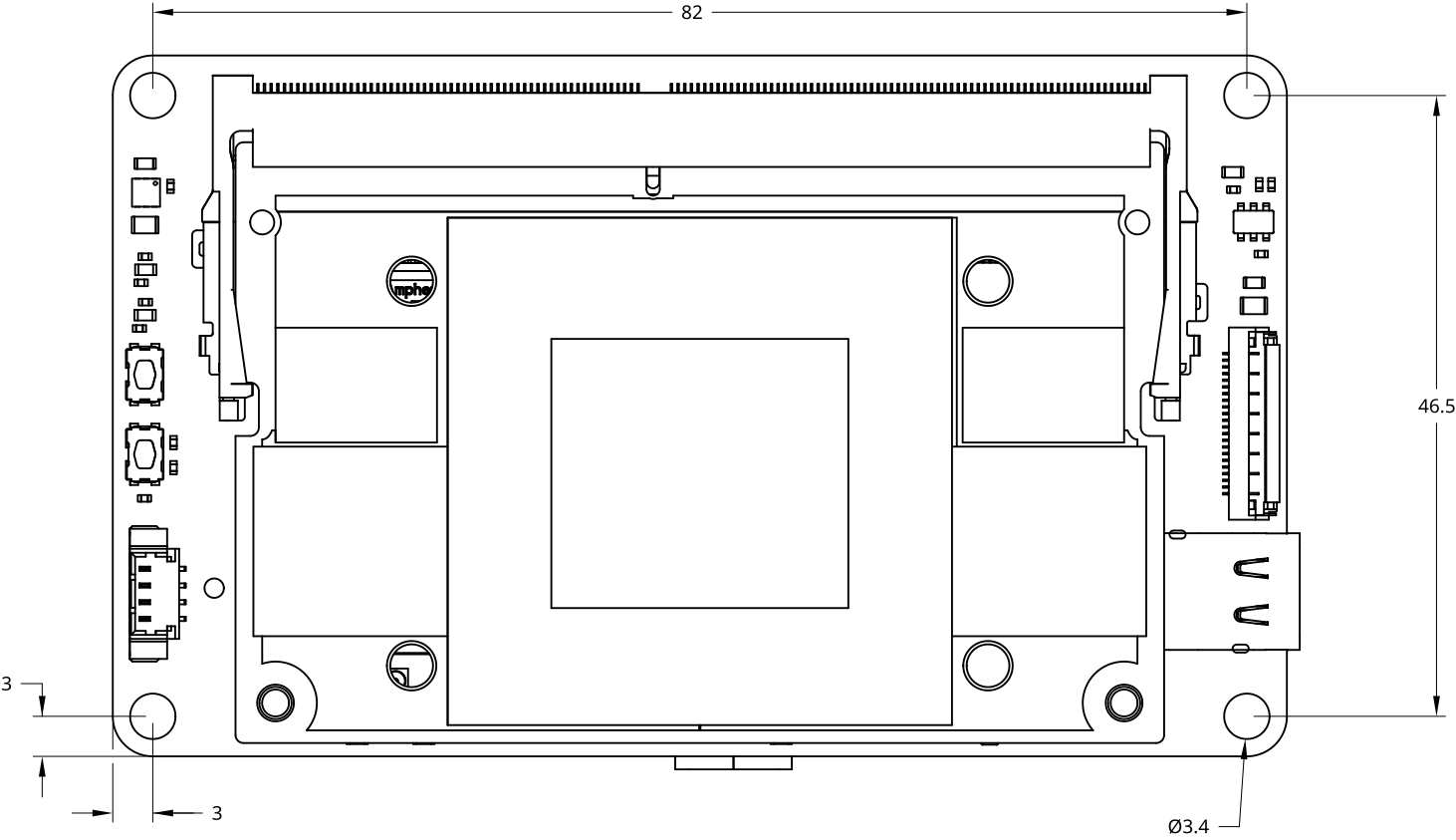


Figure 2: Aleph Carrier Board Top View

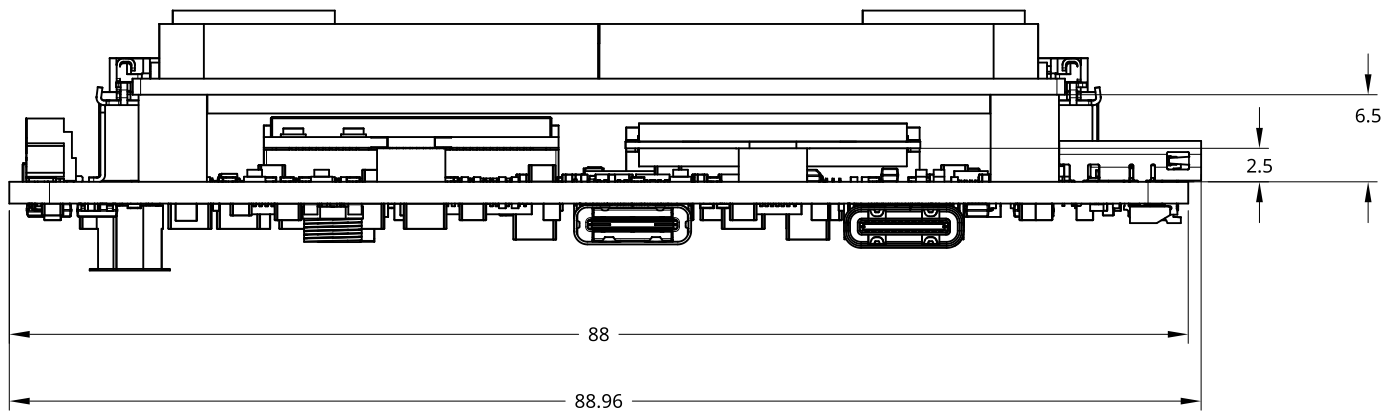


Figure 3: Aleph Carrier Board Front View

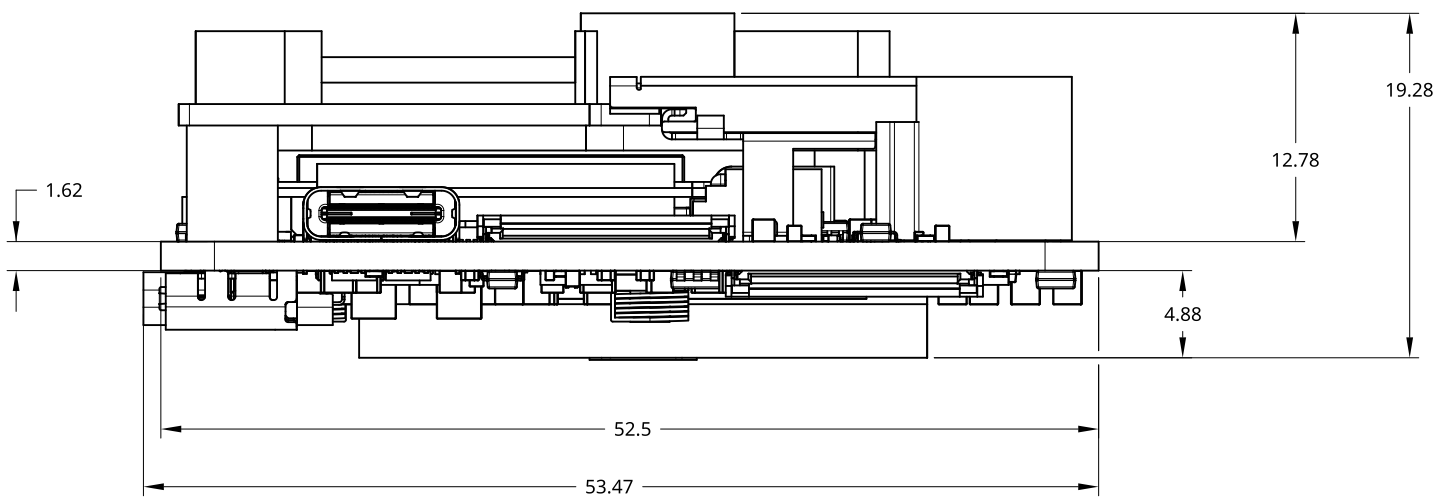


Figure 4: Aleph Carrier Board Side View

Electrical and Hardware

Minimum operating supply voltage (V)	5
Maximum operating supply voltage (V)	20
Power consumption (W)	Idle: 10 Peak: 40

Buttons

Designator	Function
SW2	Force Recovery
SW3	Reset

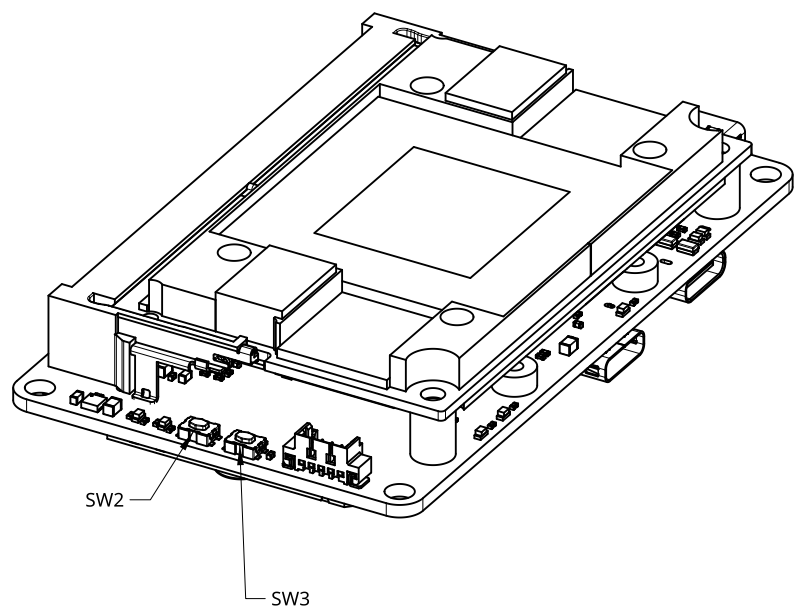


Figure 5: Aleph Carrier Board Buttons Callout

Connectivity

Designator	Connector Type	Interface
J1, J2	2x CSI-2 Interface	<u>22-pin MIPI</u> , similar to Raspberry Pi 5
J3	<u>Samtech ERF5-050 B2B connector</u>	• 2x PCIe 4.0 x1 • USB-SS (5 Gbps) • 2x I2C (VDDIO 3.3 V) • SPI (VDDIO 1.8 V) • CAN TTL (VDDIO 3.3 V) • UART (VDDIO 1.8 V) • 6x GPIO (VDDIO 1.8 V) • GigE
J4	MOLEX_0533980471 (4-pin)	Jetson fan connector
J5	260 Position SODIMM	SOM connector
J6	M.2 E Key	• PCIe 4.0 x1 • USB HS (480 Mbps) • I2C • UART
J7	M.2 M Key	PCIe 4.0 x4
J8	USB-C	Serial Debug Port with USB Power Delivery
J9	USB-C	USB-SS+ (10 Gbps) with USB Power Delivery and DisplayPort Alt mode
J10	USB-C	USB-SS+ (10 Gbps)

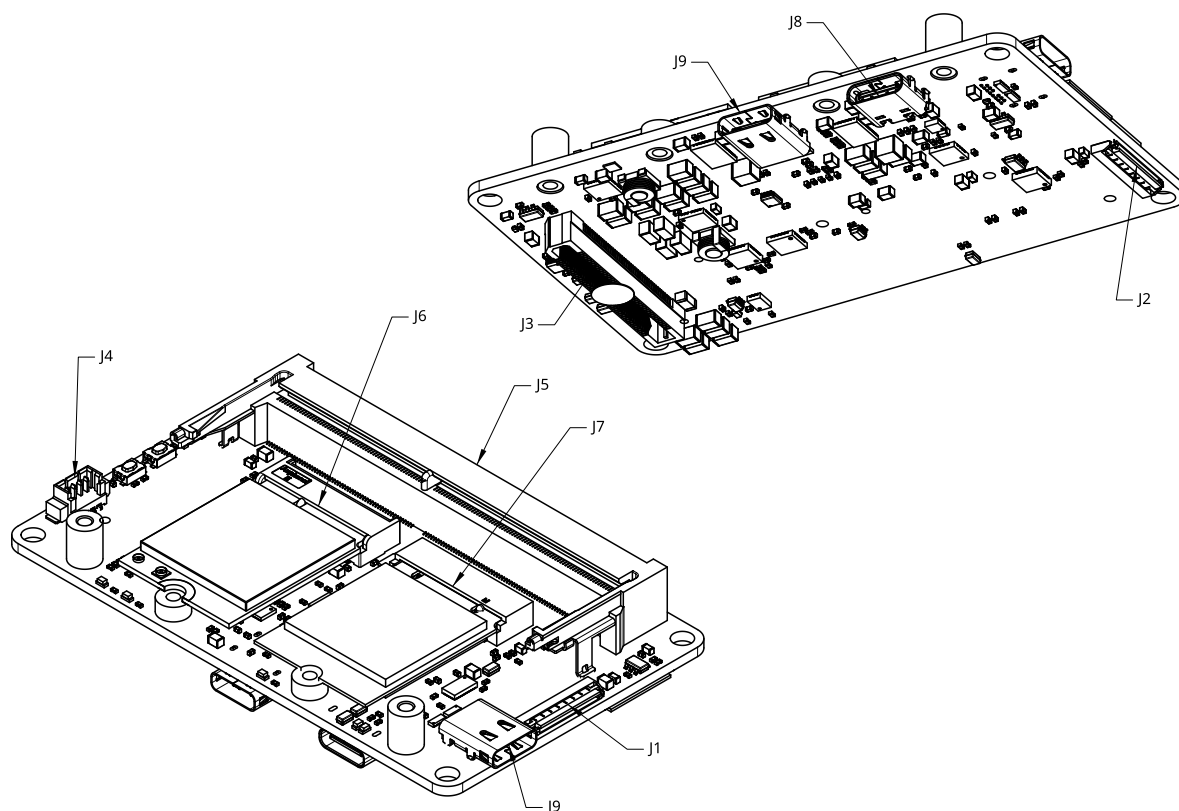


Figure 6: Connector Callouts

Board to Board Connector Pinout

Pin-1	GND	Pin-26	PCIE3-CLKREQ	Pin-51	GBE-MDI2_N	Pin-76	CAN-RX
Pin-2	GND	Pin-27	GND	Pin-52	GND	Pin-77	I2C1-SDA
Pin-3	PE3T0_N	Pin-28	PCIE-WAKE	Pin-53	GND	Pin-78	CAN-TX
Pin-4	GND	Pin-29	GPI01	Pin-54	GBE-LED-ACT	Pin-79	I2C1-SCL
Pin-5	PE3T0_P	Pin-30	GND	Pin-55	GBE-LED-LINK	Pin-80	GND
Pin-6	PCIE2-RX0_N	Pin-31	GPI07	Pin-56	GND	Pin-81	GND
Pin-7	GND	Pin-32	GPI013	Pin-57	GND	Pin-82	PMIC_BBAT
Pin-8	PCIE2-RX0_P	Pin-33	GPI09	Pin-58	EXP-USB1-D_P	Pin-83	SPARE0
Pin-9	PE2T0_N	Pin-34	GPI012	Pin-59	SPI0-SCK	Pin-84	SPARE1
Pin-10	GND	Pin-35	GND	Pin-60	EXP-USB1-D_N	Pin-85	GND
Pin-11	PE2T0_P	Pin-36	GPI011	Pin-61	SPI0-MOSI	Pin-86	EXP-VDD
Pin-12	PCIE2/3-RX1_N	Pin-37	UART1-RXD	Pin-62	GND	Pin-87	GND
Pin-13	GND	Pin-38	GND	Pin-63	SPI0-MISO	Pin-88	EXP-VDD
Pin-14	PCIE2/3-RX1_P						

Pin-15	PCIE2-CLK_N	Pin-39	UART1-RXD	Pin-64	EXP-USB1-D_N	Pin-89	GND
Pin-16	GND	Pin-40	GND	Pin-65	SPI0-CS0	Pin-90	EXP-VDD
Pin-17	PCIE2-CLK_P	Pin-41	GND	Pin-66	USBSS1-RX_P	Pin-91	GND
Pin-18	PCIE3-CLK_N	Pin-42	GBE-MDI1_P	Pin-67	SPI0-CS1	Pin-92	EXP-VDD
Pin-19	GND	Pin-43	GBE-MDI0_P	Pin-68	GND	Pin-93	GND
Pin-20	PCIE3-CLK_P	Pin-44	GBE-MDI1_N	Pin-69	GND	Pin-94	EXP-VDD
Pin-21	PCIE2-RST	Pin-45	GBE-MDI0_N	Pin-70	USBSS1-TX_N	Pin-95	GND
Pin-22	GND	Pin-46	GND	Pin-71	I2C0-SDA	Pin-96	EXP-VDD
Pin-23	PCIE2-CLKREQ	Pin-47	GND	Pin-72	USBSS1-TX_P	Pin-97	GND
Pin-24	PCIE3-CLK_P	Pin-48	GBE-MDI3_P	Pin-73	I2C0-SCL	Pin-98	EXP-VDD
Pin-25	GND	Pin-49	GBE-MDI2_P	Pin-74	GND	Pin-99	GND
		Pin-50	GBE-MDI3_N	Pin-75	GND	Pin-100	EXP-VDD

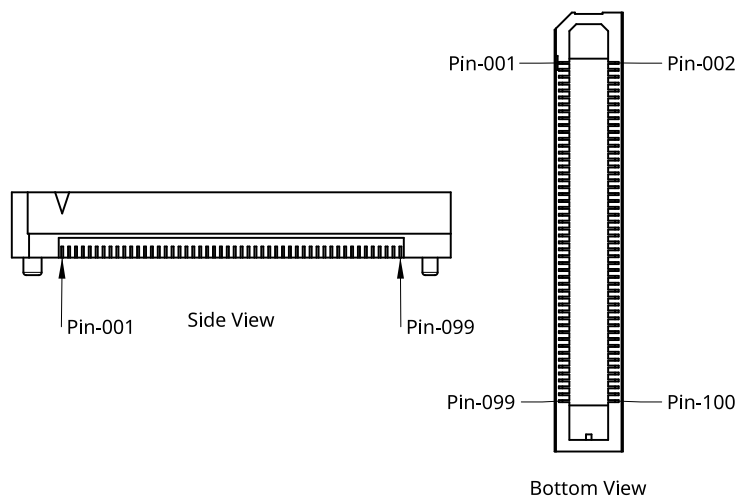


Figure 7: Board to Board Connector Reference

Compatible SOMs

The Aleph Carrier Board supports all NVIDIA Jetson Orin NX series SOMs (8GB and 16GB variants) and the Jetson Orin Nano series SOMs. The board interfaces and houses the some using a 260 position SODIMM connector. It provides full electrical and mechanical compatibility per NVIDIA design guidelines.

SOM	Jetson Orin NX 8GB	Jetson Orin NX 16GB	Jetson Orin Nano 8GB
AI performance (Sparse INT8 TOPS)	117	157	67
GPU	1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores	1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores	1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores
GPU max frequency (MHz)	1173	1173	1020
CPU	6-core Arm® Cortex®-A78AE v8.2 64-bit CPU 1.5MB L2 + 4MB L3	8-core Arm® Cortex®-A78AE v8.2 64-bit CPU 2MB L2 + 4MB L3	6-core Arm® Cortex®-A78AE v8.2 64-bit CPU 1.5MB L2 + 4MB L3
CPU max frequency (GHz)	2	2	1.7
DL accelerator	1x NVDLA v2	2x NVDLA v2	-
DLA max frequency (GHz)	1.23	1.23	-
Memory	8GB 128-bit LPDDR5 102.4GB/s	16GB 128-bit LPDDR5 102.4GB/s	8GB 128-bit LPDDR5 102 GB/s
Storage	Supports external NVMe	Supports external NVMe	Supports external NVMe
Vision Accelerator	1x PVA v2	1x PVA v2	-

Software

Operating System Support

The Aleph Carrier Board flight computer comes pre-flashed with NixOS. The system is also compatible with Jetpack and Yocto.